

Interactive Web-Controlled LED System

with Capacitive Touch using ESP32-S3

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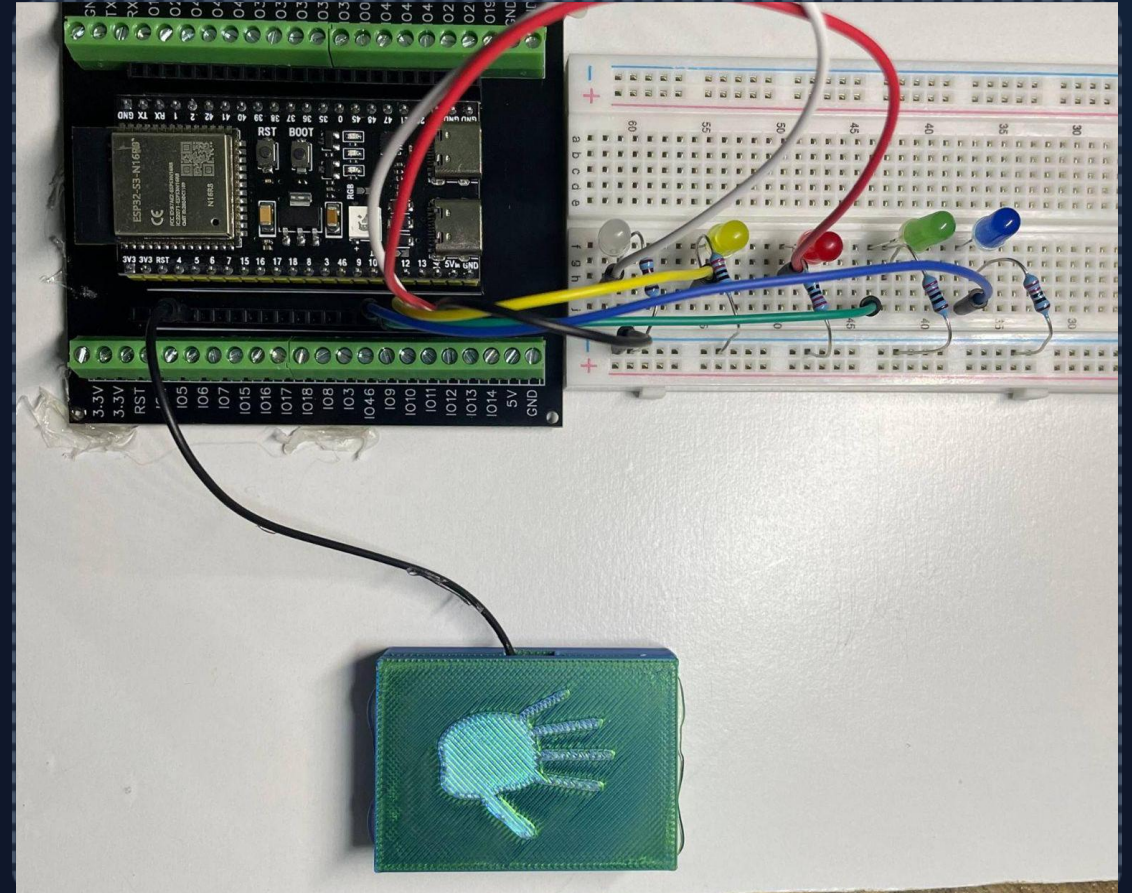
Major: Computer Engineering

Course: [Embedded Systems]

Instructor: [IDr.Faisal Alamri]

Project Overview

- **The Objective:** Integrate digital web control with physical interactions to create a seamless IoT lighting system.
- **The Core:** Powered by the highly efficient ESP32-S3 microcontroller.
- **Dual Control:** Operates via a wireless Local Web Server and a physical hidden capacitive touch sensor.
- **Performance:** Achieves zero-latency response for both web commands and physical touch inputs.



Hardware Architecture



Microcontroller

ESP32-S3 Dev Module attached to a breakout board. Selected for its built-in Wi-Fi and native capacitive touch support.



LED Indicators

Five LEDs (White, Yellow, Red, Green, Blue) connected via specific digital output pins (13, 12, 14, 11, 10).

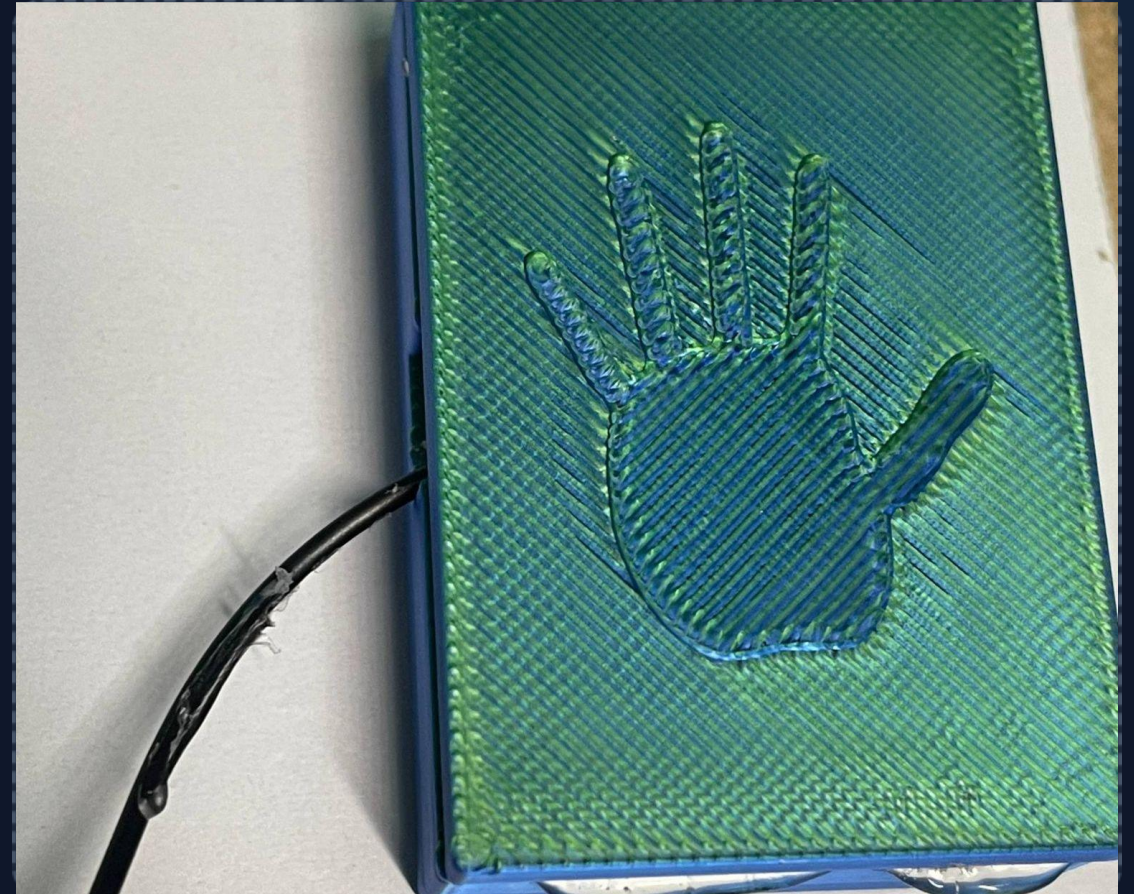


Common Ground

A Breadboard integrates the Ground (GND) lines for all LEDs, ensuring electrical stability and safe current flow.

Touch Sensor System

- > **The Innovation:** A custom capacitive sensing field created using highly conductive **Nickel Strips**.
- > **Pin Configuration:** Connected to designated touch pin IO4 on the ESP32-S3.
- > **Placement:** Attached directly to the inner roof of the enclosure.
- > **Efficiency:** The Nickel strips concentrate the electromagnetic field, enabling precise detection through 5mm of solid 3D-printed plastic.



Software Implementation

Local Web Server

The ESP32-S3 hosts a lightweight HTML page on the local Wi-Fi network.

- > Processes HTTP Requests to toggle LEDs.
- > Supports individual color control.
- > Provides a "Turn All ON/OFF" master override.
- > Maintains perfect synchronization of LED states.

Touch Algorithm

A continuous loop reads capacitance values and classifies physical interactions:

- > **Short Press:** Cycles to the next LED color.
- > **Double Press:** Adds the next color to the active lights.
- > **Long Press:** Master reset; turns off all LEDs instantly.

Engineering Challenges



Ghost Touches

Issue: High sensitivity caused random LED triggers.

Solution: Analyzed raw capacitance via Serial Monitor.
Identified idle (25,600) vs. touch (28,000) values.
Calibrated threshold perfectly to 27000.



Insulation Barrier

Issue: Aluminum foil and adhesive tape created an insulator, dropping detection range.

Solution: Swapped to **Nickel Strips** to concentrate the EM field, solving the issue entirely.

Final Prototype

The completely assembled Interactive LED System.

- > Compact wiring hidden inside the 3D-printed shell.
- > Zero latency between web input and LED response.
- > Invisible touch interface maintaining a clean exterior design.



Future Enhancements



Cloud IoT

Integrating platforms like Blynk to transition from a local server to global remote control via mobile apps.



OLED Display

Adding a small I2C OLED screen to dynamically display the assigned IP Address and real-time LED statuses.



Auto-Brightness

Implementing a Light Dependent Resistor (LDR) to automatically adjust LED intensity based on room lighting.